

4.13.86

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Question

A plane which is perpendicular to two planes

$$2x - 2y + z = 0 \quad \text{and} \quad x - y + 2z = 4 \quad (1)$$

passes through $(1, -2, 1)$. The distance of the plane from the point $(1, 2, 2)$ is (2006).

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

Given two planes

$$\mathbf{n}_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \quad \mathbf{n}_2 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad \mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \quad (2)$$

Let $\mathbf{n}$ be normal vector required plane

$$\mathbf{n}_1^T \mathbf{n} = 0 \quad (3)$$

$$\mathbf{n}_2^T \mathbf{n} = 0 \quad (4)$$

$$\begin{pmatrix} \mathbf{n}_1 & \mathbf{n}_2 \end{pmatrix}^T \mathbf{n} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (5)$$

Theoretical Solution

From solving we

$$\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad (6)$$

Equation of plane is

$$\mathbf{n}^T \mathbf{x} = c \quad (7)$$

$$\mathbf{n}^T \mathbf{a} = c = -1 \quad (8)$$

distance of point from plane is given by

$$k = \frac{|\mathbf{n}^T \mathbf{x} - c|}{\|\mathbf{n}\|} \quad (9)$$

$$k = \frac{|\mathbf{n}^T \mathbf{b} - c|}{\|\mathbf{n}\|} = \frac{4}{\sqrt{2}} \quad (10)$$

```
#include <stdio.h>
#include <math.h>

int main() {
    // Normals of given planes
    double n1[3] = {2, -2, 1};
    double n2[3] = {1, -1, 2};

    // Cross product to get required plane normal
    double a = n1[1]*n2[2] - n1[2]*n2[1];
    double b = n1[2]*n2[0] - n1[0]*n2[2];
    double c = n1[0]*n2[1] - n1[1]*n2[0];

    // Point through which plane passes
    double x0 = 1, y0 = -2, z0 = 1;
```

```
// Plane equation:  $a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$ 
double d = -(a*x0 + b*y0 + c*z0);

printf("Equation of required plane: %.1fx + %.1fy + %.1fz +
      %.1f = 0\n", a, b, c, d);

// Point from which distance is required
double x1 = 1, y1 = 2, z1 = 2;

// Distance formula
double numerator = fabs(a*x1 + b*y1 + c*z1 + d);
double denominator = sqrt(a*a + b*b + c*c);
double distance = numerator / denominator;

printf("Distance = %.4f\n", distance);

return 0;
}
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Plane:  $x + y + 1 = 0 \Rightarrow \text{normal} = (1, 1, 0)$ 
a, b, c, d = 1, 1, 0, 1 #  $ax + by + cz + d = 0$ 

# Point
P = np.array([1, 2, 2])

# Distance formula
dist = abs(a*P[0] + b*P[1] + c*P[2] + d) / np.sqrt(a*a + b*b + c*c)
print("Distance =", dist)

# Foot of perpendicular
t = -(a*P[0] + b*P[1] + c*P[2] + d) / (a*a + b*b + c*c)
foot = P + t * np.array([a, b, c])
```

Python Code

```
# Grid for plane
x = np.linspace(-5, 5, 20)
y = np.linspace(-5, 5, 20)
X, Y = np.meshgrid(x, y)
Z = np.zeros_like(X) # since c=0, plane is vertical (no z-
    dependence)

# Plot
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

# Highlighted Plane (solid color + edge lines)
ax.plot_surface(X, Y, Z, color='lightblue', alpha=0.8, edgecolor=
    'k', linewidth=0.5)

# Point
ax.scatter(P[0], P[1], P[2], color='red', s=60, label=f"Point {
    tuple(P)}")
```


Python Code

```
# Foot of perpendicular
ax.scatter(foot[0], foot[1], foot[2], color='green', s=60, label=
    "Foot of perpendicular")

# Distance line
ax.plot([P[0], foot[0]], [P[1], foot[1]], [P[2], foot[2]], 'k--',
    lw=2, label=f"Distance = {dist:.2f}")

# Labels
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")
ax.set_title("Highlighted Plane with Point-to-Plane Distance")

ax.legend()

# Save
plt.savefig("highlighted_plane_distance.png", dpi=300)
plt.show()
```

Highlighted Plane with Point-to-Plane Distance

